Population-scale discovery of tumor driving genefusions

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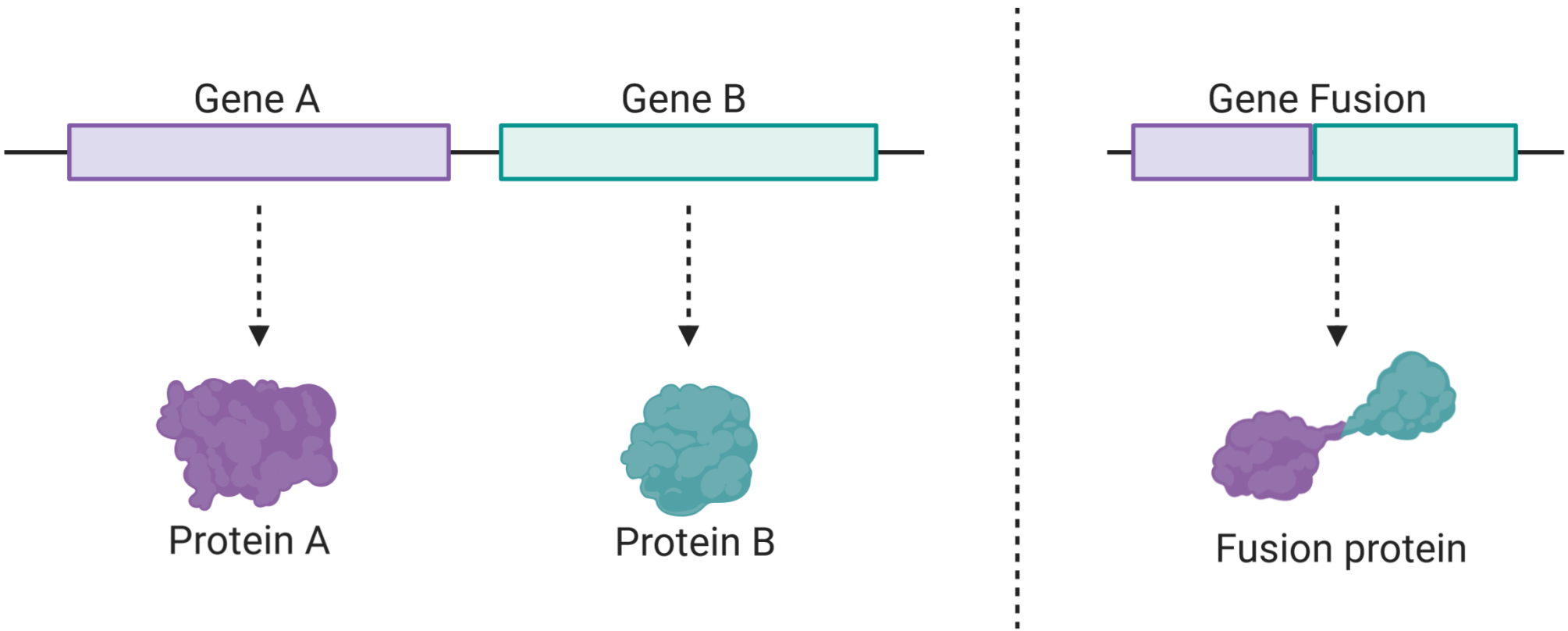
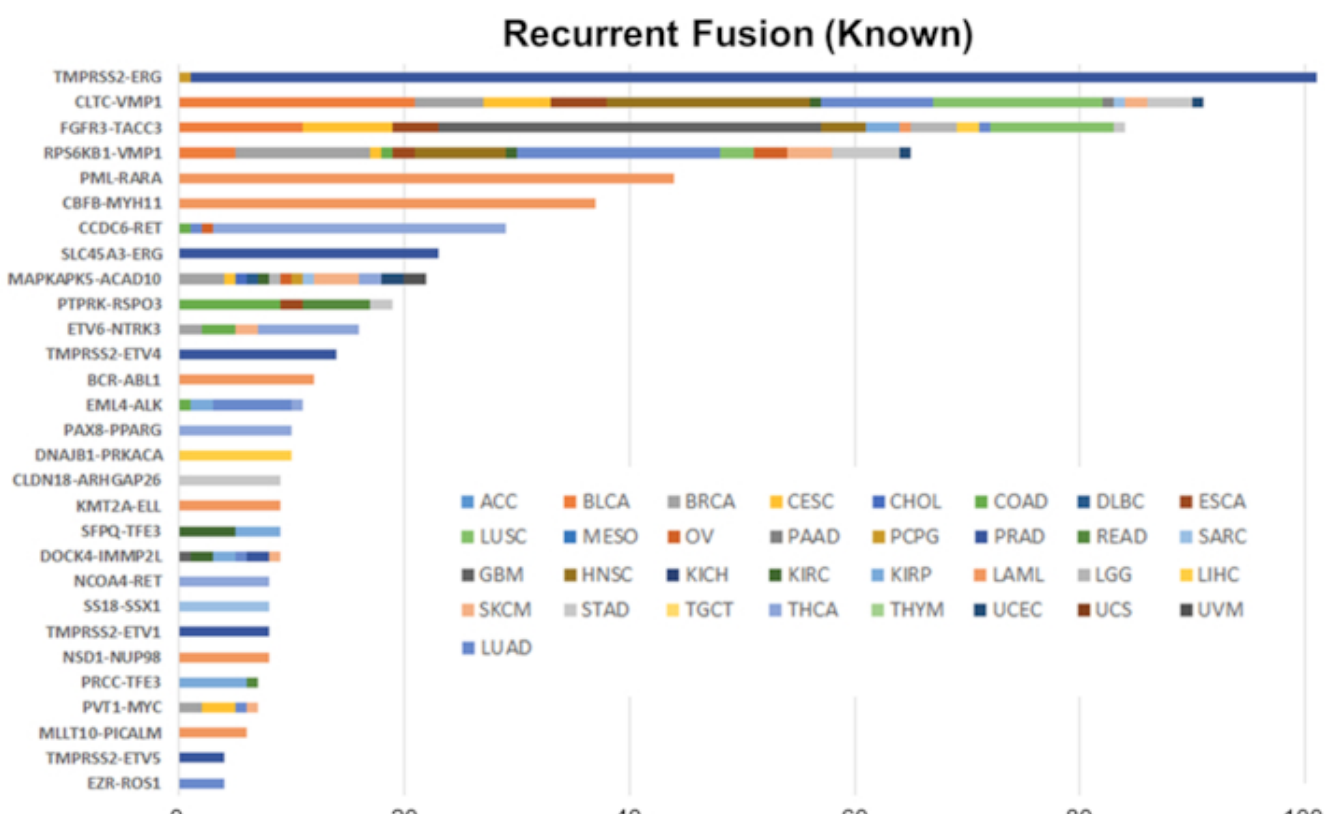
Genefusions

- Genefusions are mutations joining two genes together



Recurrent tumor fusions

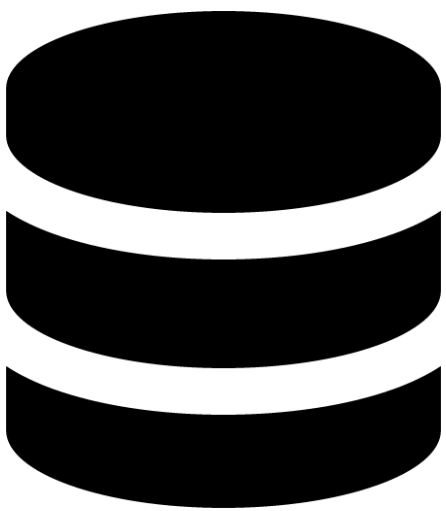
- In human cancer, some fusions are known to drive tumor progression
- **Recurrent** fusions are potential diagnostic and therapeutic targets



E.g., ERG-TMPRSS2 is a diagnostic marker, occuring in ~50% of prostate cancer cases

Genomic interval databases

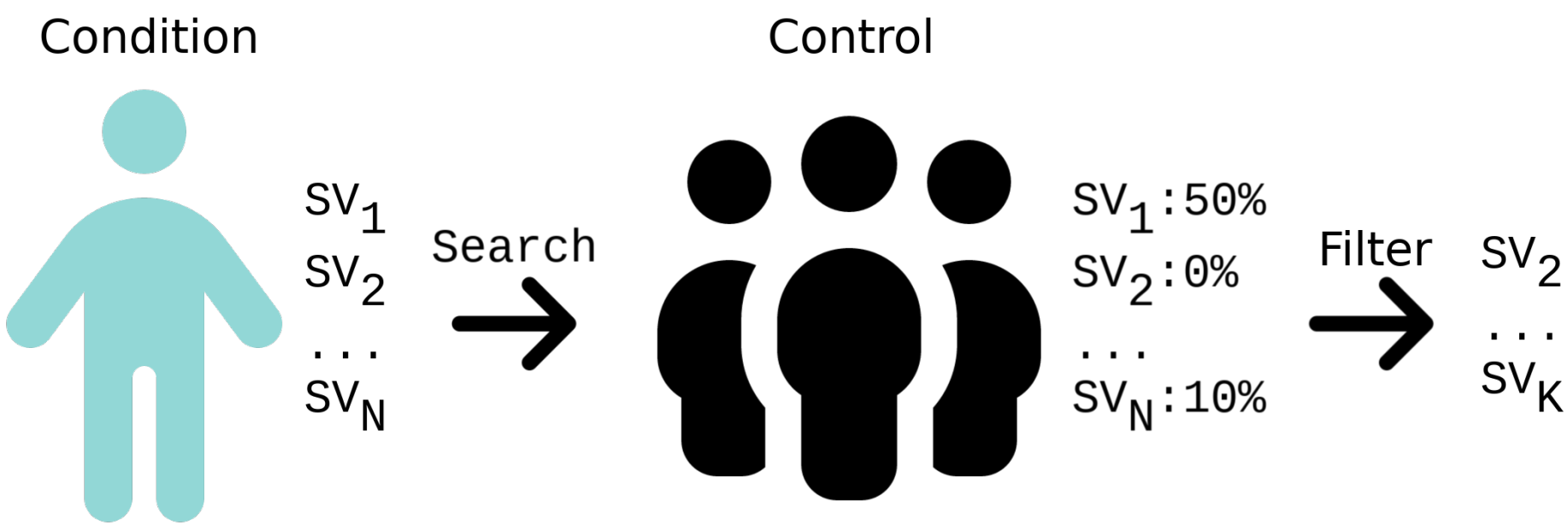
- Storage and retrieval of fusions across populations is enabled by genomic interval databases (STIX & GIGGLE)



STIX & GIGGLE developed by Layer Lab

Adjacent work

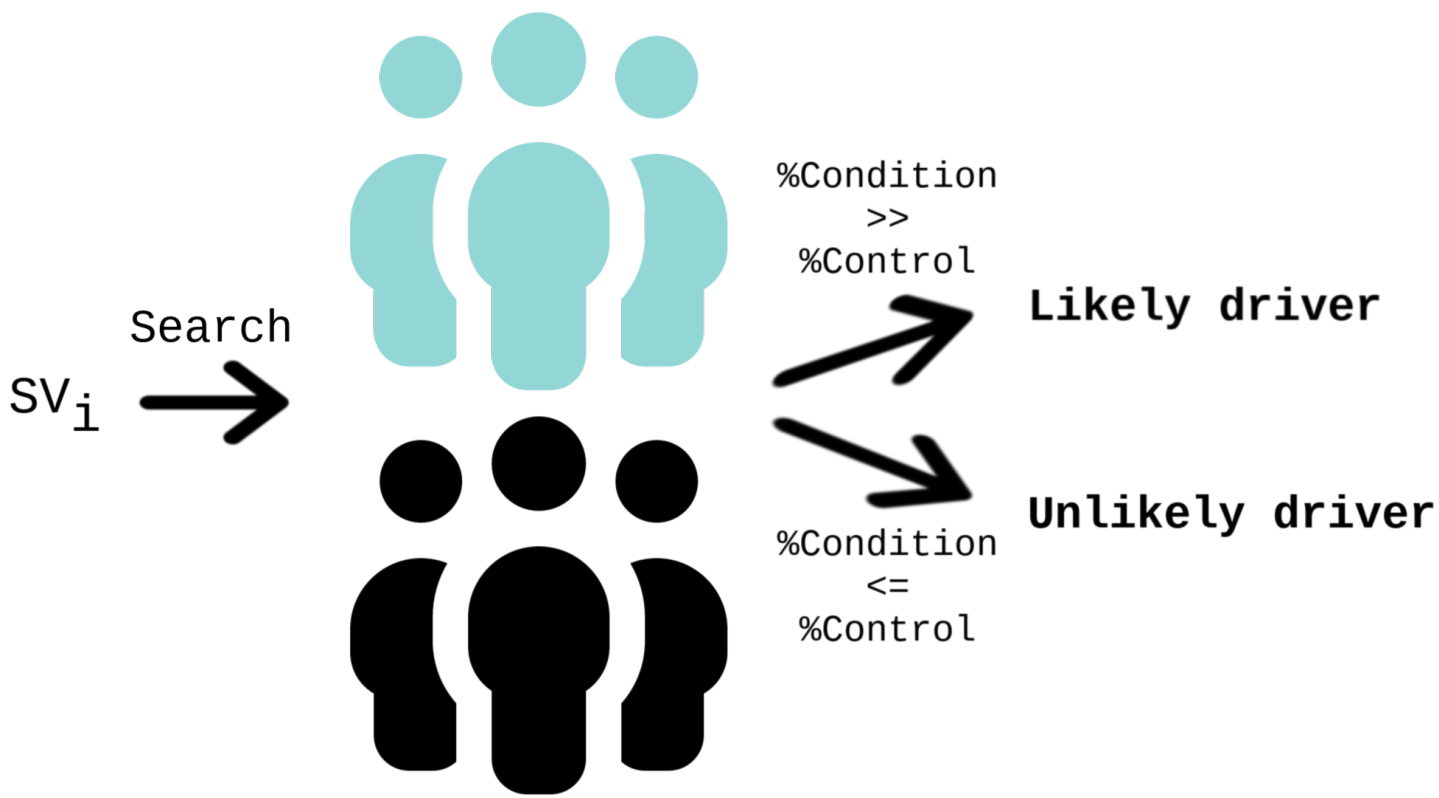
- Population databases are already used for *filtering* unlikely tumor-driving structural variants (SV)s



Fusions are a special case of SVs

Current approach

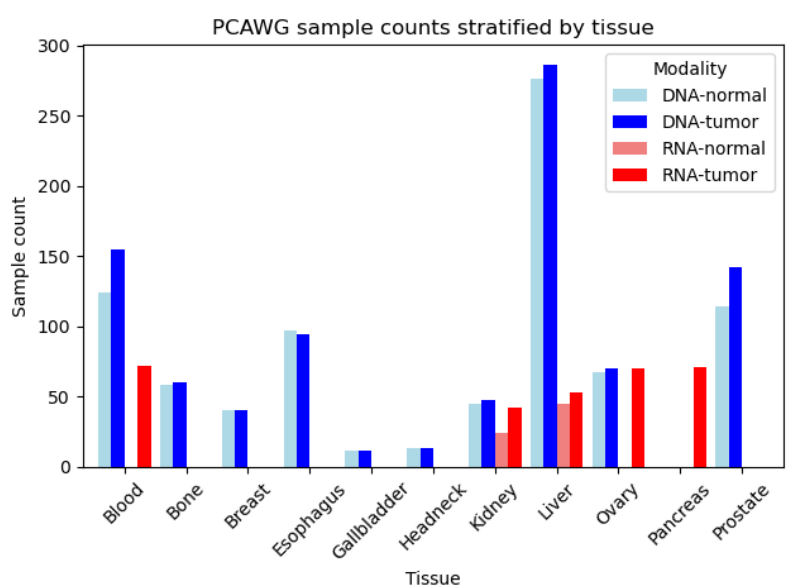
- Here, population databases are used for *discovery* of tumor-driving fusions



Need a large cancer cohort to find recurrent fusions

Data

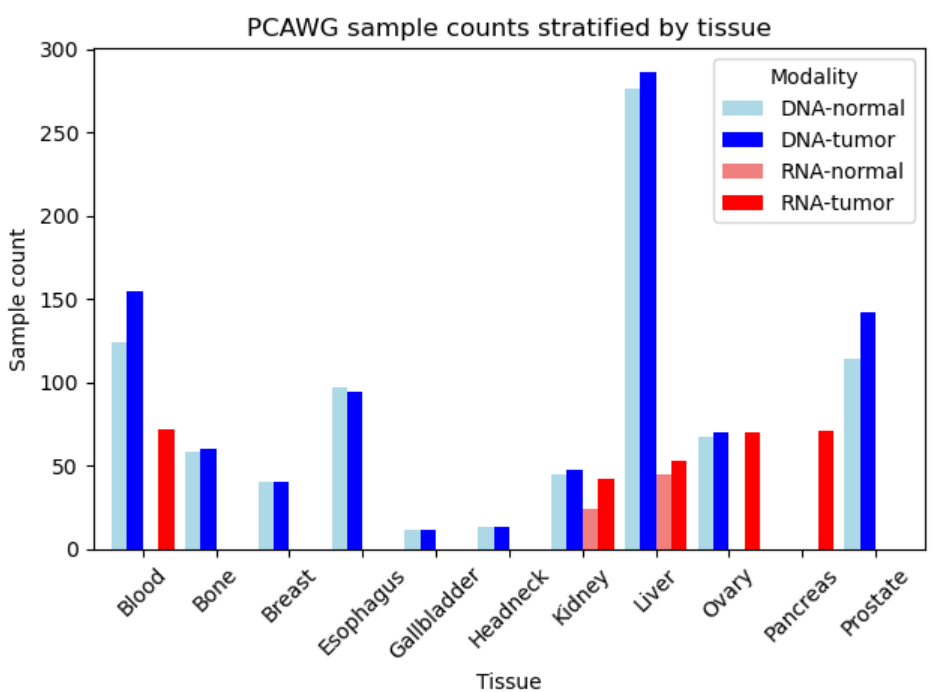
- Tumor and control (WGS & RNA-seq): 2,143 Pan-cancer Analysis of Whole Genomes (PCAWG) samples
- Control (WGS): 2,504 samples from 1000G project
- Total 4,647 human genome samples



Variability in tissue, sequencing technology, and disease state across samples

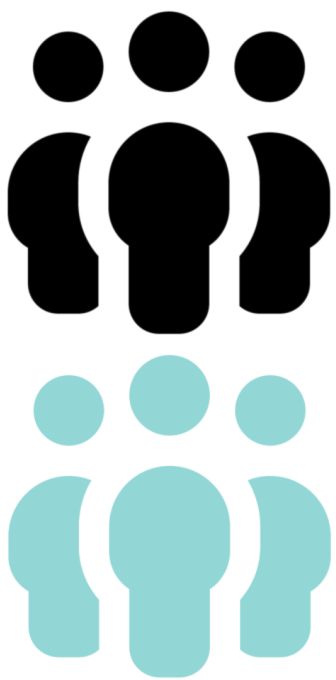
Fusion prioritization

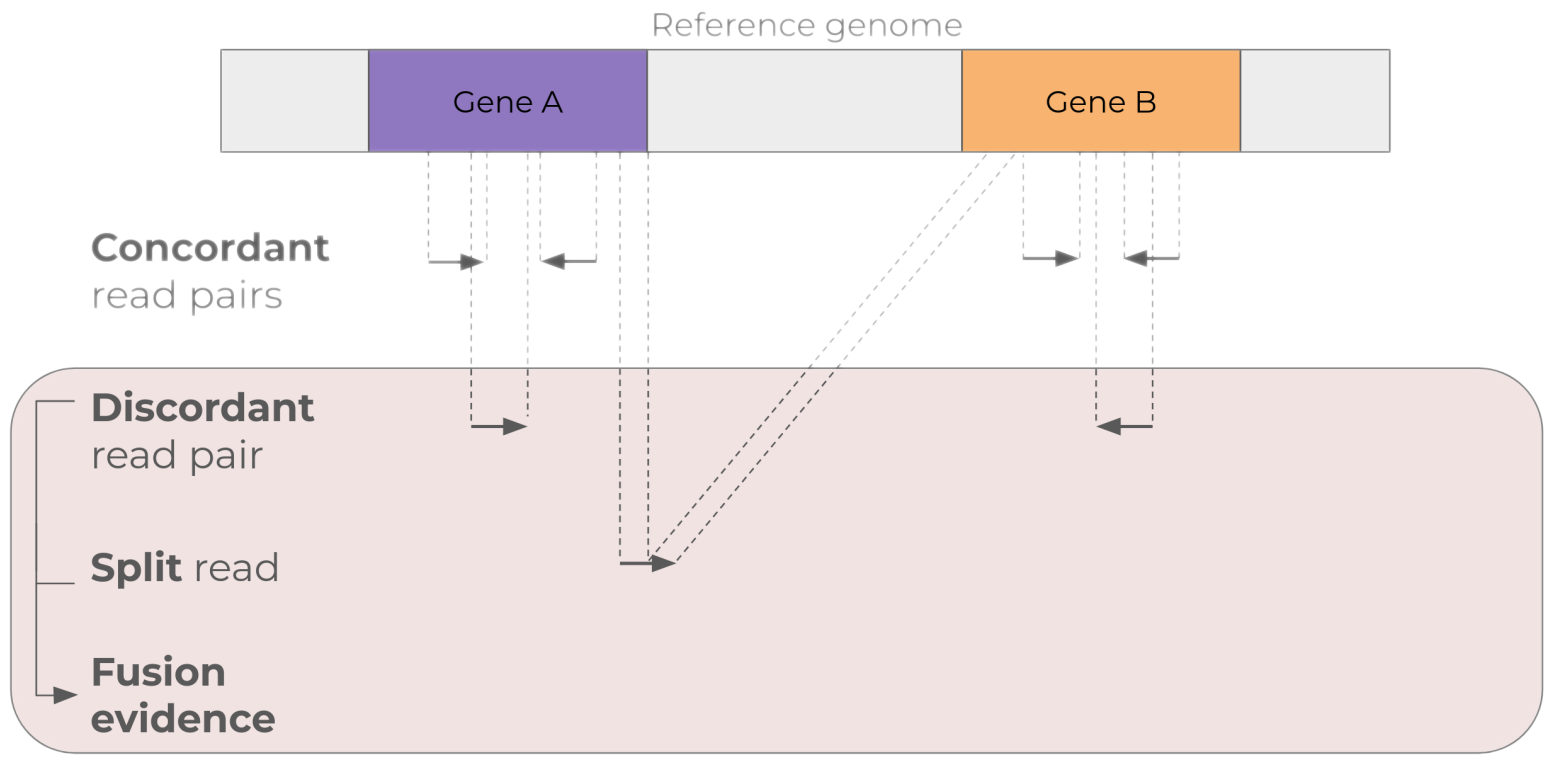
- How prioritize most likely tumor-driving fusions given the multimodal evidence ($\mathbf{x}$)?



$$f(\mathbf{x}) = ?$$

Fundamental units of fusion evidence

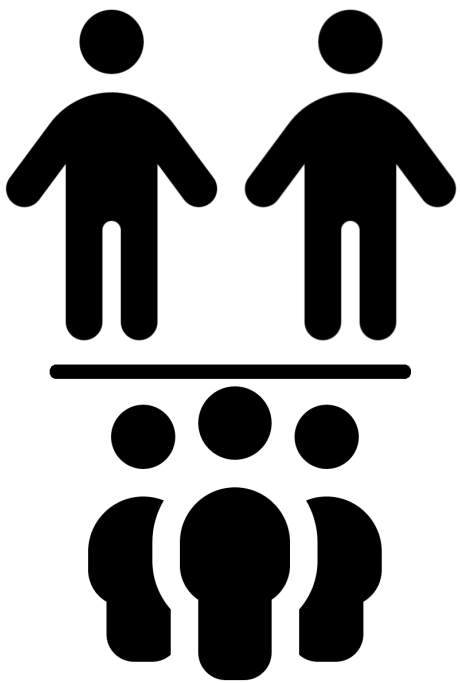


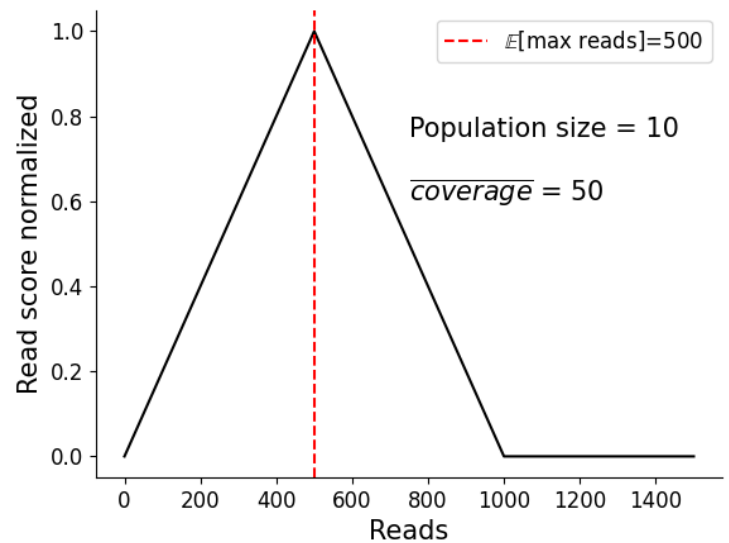


$$\text{score} = \#\text{reads} + \#\text{samples} - \#\text{reads} - \#\text{samples}$$

Normalization

- Normalization accounts for population size and sequencing parameters (coverage)





$$\text{score}_{\text{sample}} \in [0,1] = \frac{\#\text{samples}}{\text{Population size}} \quad \text{score}_{\text{reads}} \in [0,1] = \begin{cases} \frac{\mathbb{E}[\text{max reads}] - |\#\text{reads} - \mathbb{E}[\text{max reads}]|}{\mathbb{E}[\text{max reads}]} & \text{if } \#\text{reads} \leq 2 \cdot \mathbb{E}[\text{max reads}] \\ 0 & \text{else} \end{cases}$$

Score sign interpretation

- score $\in \mathbb{R}_{++} \rightarrow$ tumor evidence > normal evidence

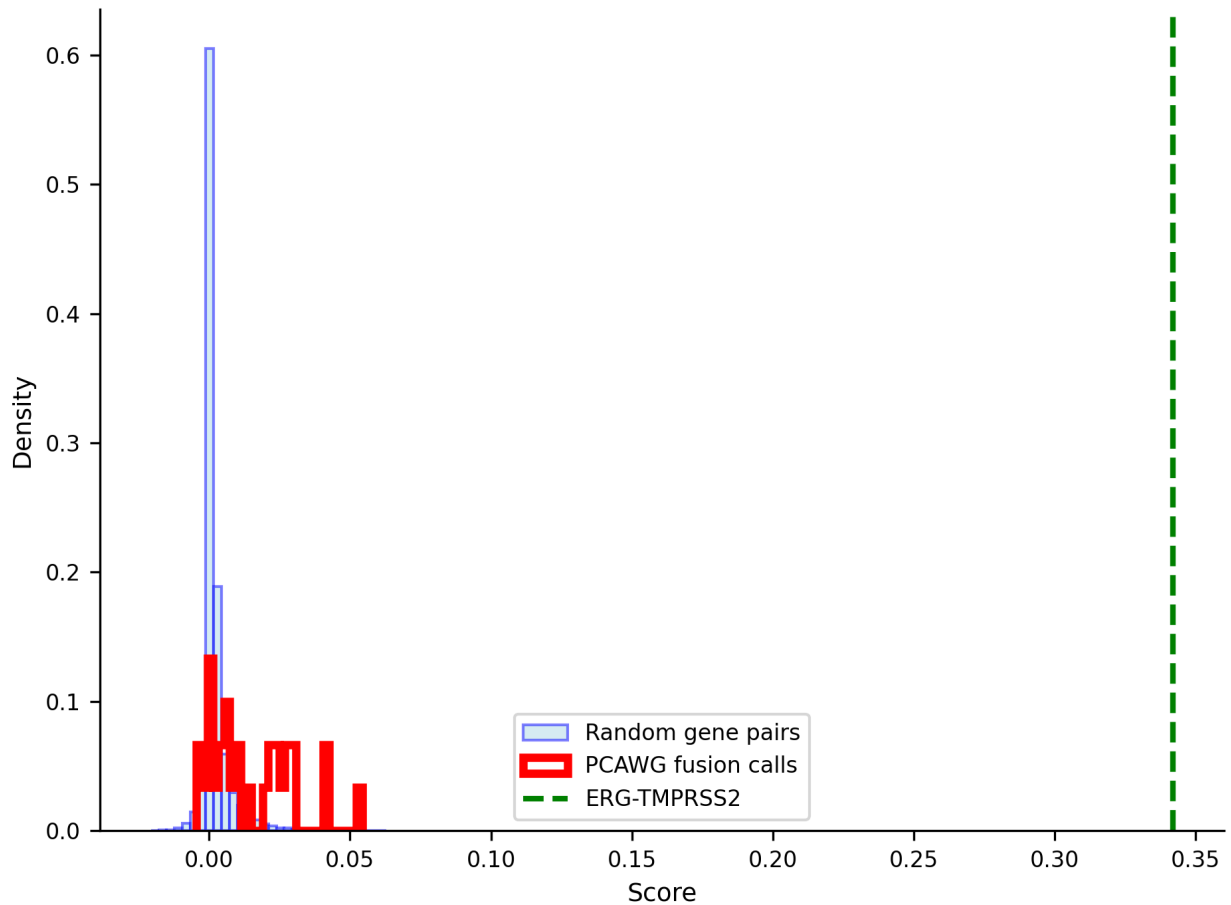
$$\text{score} = \frac{1}{2}(\text{score}_{\text{reads}} + \text{score}_{\text{samples}} - \text{score}_{\text{reads}} - \text{score}_{\text{samples}})$$

$\frac{1}{2}$ scales score to [-1,1] range

Optional mixing parameters: $W_{normal} \in [0, 1]$ and $W_{tumor} = 1 - W_{normal}$

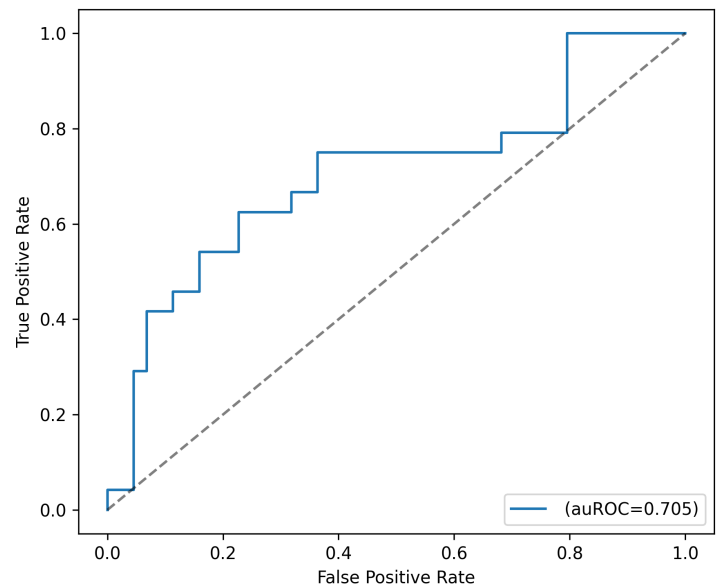
Prostate ERG-TMPRSS2 evaluation

- Recurrent tumor fusion score is higher than random gene pairs
- PCAWG prostate fusion call (not recurrent) scores are right-skewed



Recurrent normal vs. recurrent tumor classification

- Score threshold separated recurrent tumor fusions from recurrent normal fusions



Article | Open access | Published: 05 February 2020

Genomic basis for RNA alterations in cancer

PCAWG Transcriptome Core Group, Claudia Calabrese, Natalie R. Davidson, Deniz Demircioğlu, Nuno A. Fonseca, Yao He, André Kahles, Kjong-Van Lehmann, Fenglin Liu, Yuichi Shiraishi, Cameron M. Soulette, Lara Urban, Liliana Greger, Siliang Li, Dongbing Liu, Marc D. Perry, Qian Xiang, Fan Zhang, Junjun Zhang, Peter Bailey, Serap Erkek, Katherine A. Hoadley, Yong Hou, Matthew R. Huska, PCAWG Transcriptome Working Group, PCAWG Consortium + Show authors

Nature **578**, 129–136 (2020) | Cite this article

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JOURNAL ARTICLE

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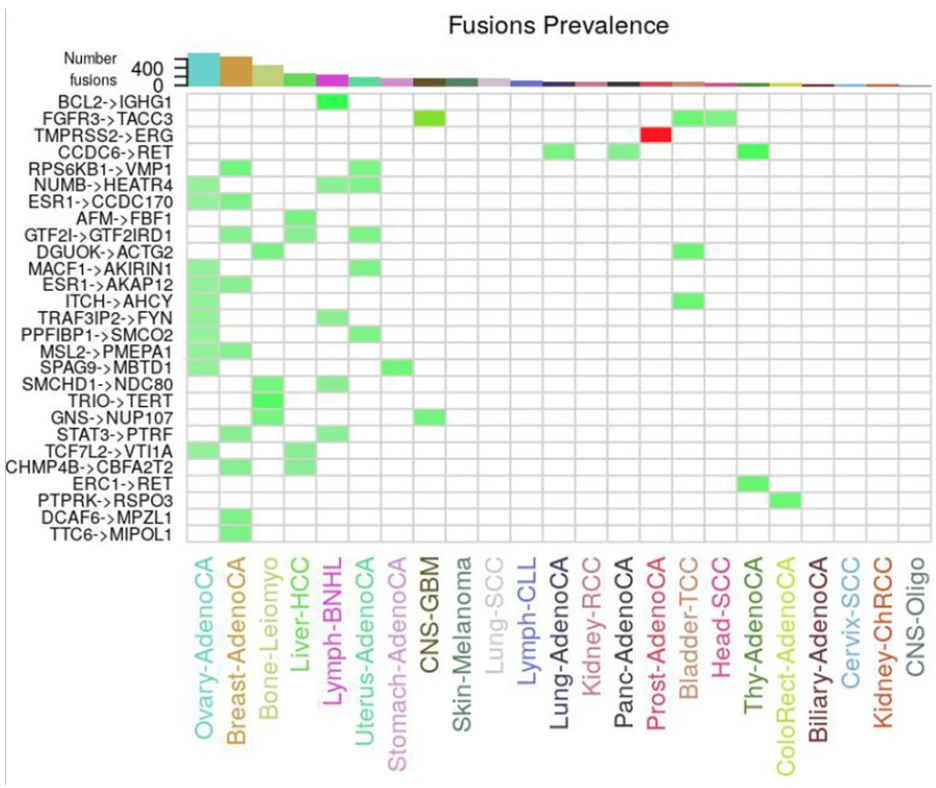
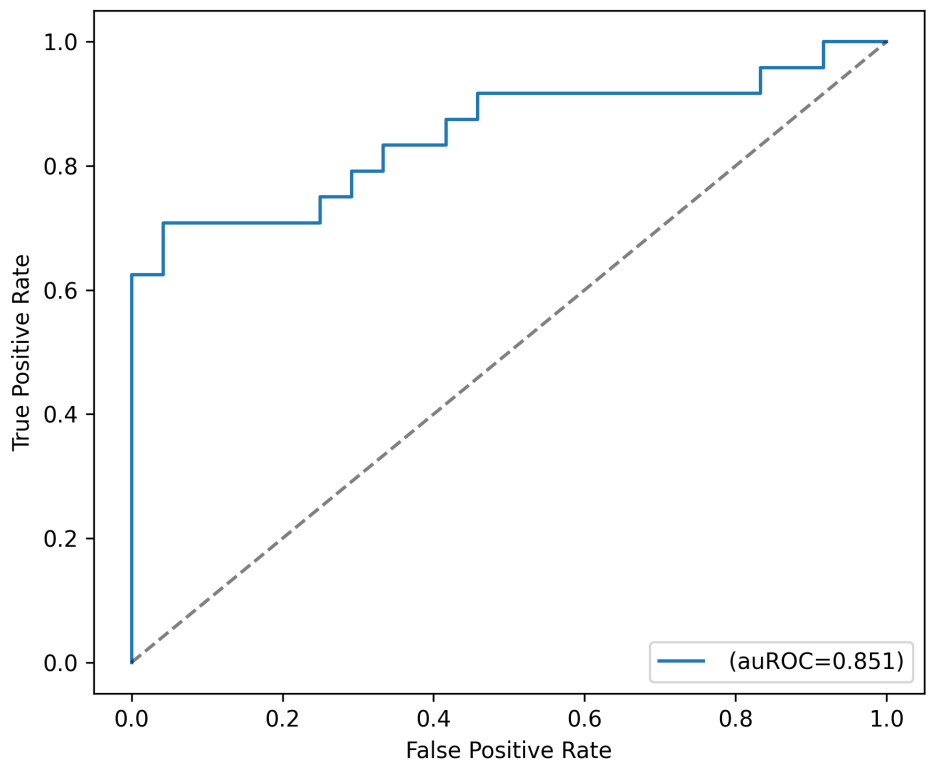
Nucleic Acids Research, Volume 44, Issue 6, 7 April 2016, Pages 2859–2872, https://doi.org/10.1093/nar/gkw032

Published: 02 February 2016 **Article history** ▾

Tissue specificity score evaluation

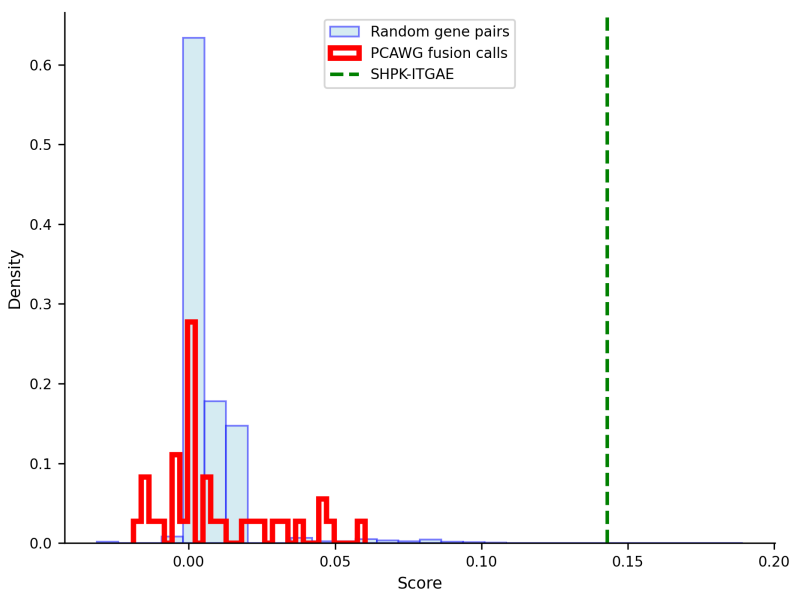
- Positives scored using “in-tissue” samples
- Negatives scored using “off-tissue” samples

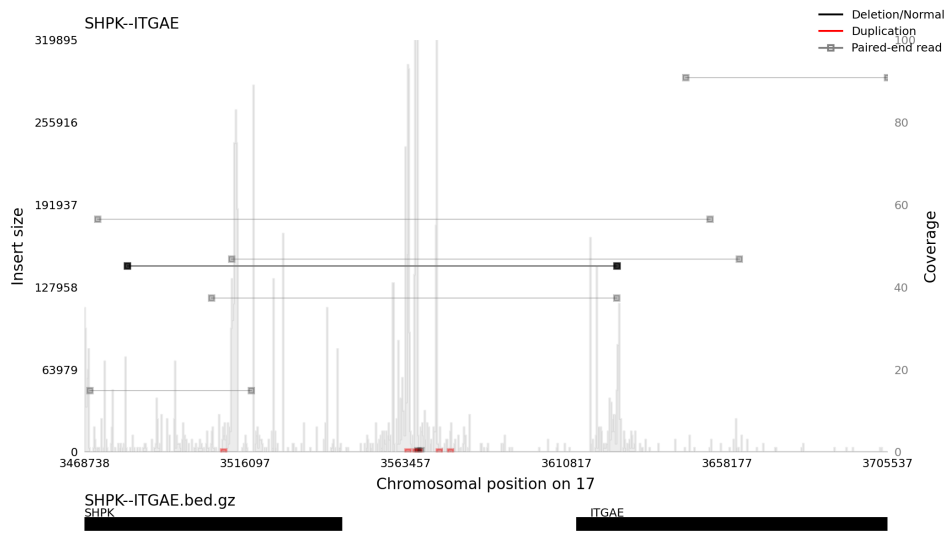




Results

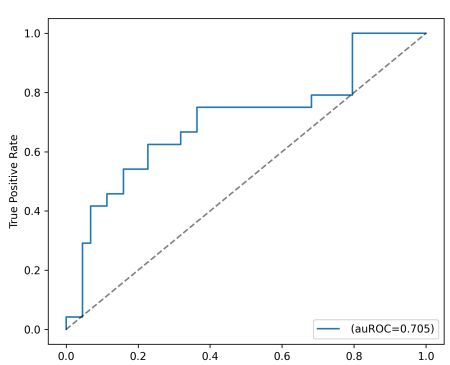
- Using score prioritization, discovered a potential novel fusion in kidney cancer: SHPK–ITGAE

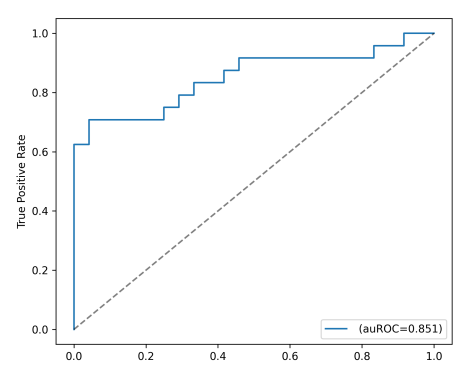


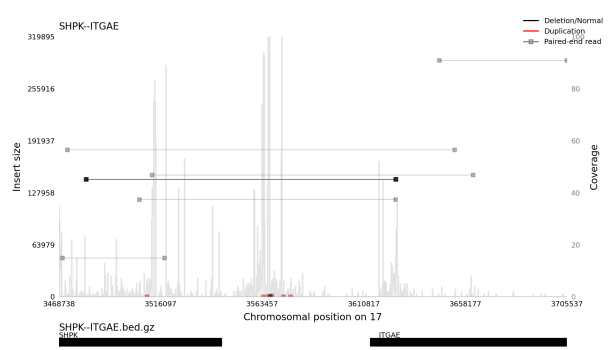


Conclusions

- Intuitive scoring collapses and ranks multimodal, multi-sample fusion evidence
- Score evaluation
  - Separation of recurrent tumor & recurrent normal fusions
  - Demonstrated tissue specificity of recurrent tumor fusions
- Potential novel fusion discovery: SHPK–ITGAE in kidney cancer
  - Not reported in primary PCAWG publication
  - Visual confirmation of read support

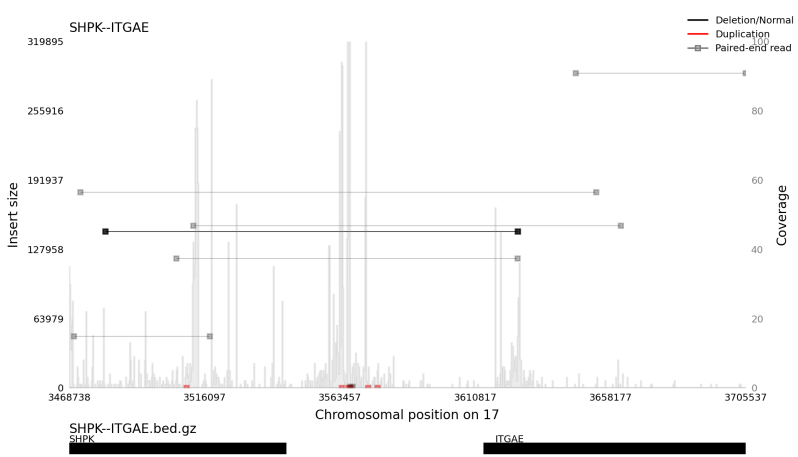


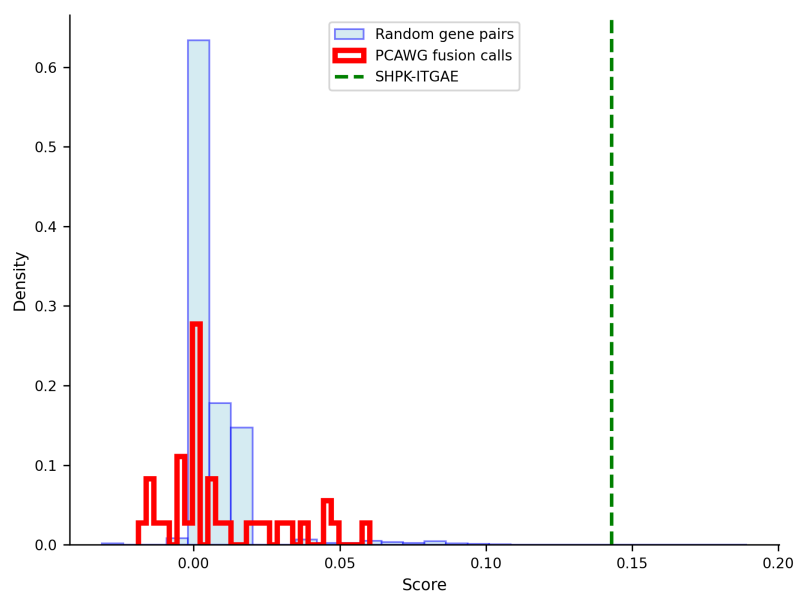




Next steps

- Potential collaboration with Anschutz Children's hospital
- More visual verification and functional analysis of putative overlooked PCAWG fusions
- Probabilistic score interpretation
  - Random gene pairs as null distribution
  - No assumption about what score values are "good/bad"





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References

- Publication and figure refernces
  - https://github.com/jakekrol/genefusion/blob/main/docs/2026_04-dbmi-symposium.qmd